EXPERIMENT: DESIGN AND CONFIGURATION OF AN INTERNETWORK USING WIRELESS ROUTER, DHCP SERVER AND INTERNET CLOUD

AIM:

To design and configure an internetwork in Cisco Packet Tracer using a wireless router, DHCP service and an Internet cloud, and to verify automatic IP address assignment and connectivity between wireless clients and the external network.

REQUIREMENTS:

Cisco Packet Tracer, one wireless router, one DHCP-capable router/server (depending on the selected topology), two wireless laptops or PCs, one server or external test host, and an Internet cloud device.

PROCEDURE:

1. Open Cisco Packet Tracer and create a new project.

2. Place a wireless router, two wireless clients, a server (if using a separate DHCP server), and an Internet cloud in the workspace.

3. Connect the wireless router's Internet/WAN port to the cloud or to the upstream network device. Connect the server to a LAN port if a separate DHCP server is used.

4. Configure the wireless router's LAN address as 192.168.10.1 with subnet mask 255.255.255.0.

5. Enable DHCP on the wireless router, or configure the separate DHCP server with an address such as 192.168.10.2 and a DHCP pool in the 192.168.10.0/24 network.

6. Set the DHCP pool to assign addresses from 192.168.10.100 onward, subnet mask 255.255.255.0, default gateway 192.168.10.1 and a DNS server address appropriate for the lab.

7. Configure the wireless SSID (for example, LAB-WIFI) and enable WPA2-PSK security with a lab password.

8. On each laptop or wireless PC, select the wireless network and set IPv4 configuration to DHCP/Automatic.

9. Check that each client receives an IP address, subnet mask, default gateway and DNS information.

10. Test connectivity by pinging the wireless router's LAN address and, where the Packet Tracer cloud/upstream network is configured to provide a reachable destination, test the external host.

11. Use Simulation mode to observe DHCP Discover, Offer, Request and Acknowledge messages where supported.

12. Save the Packet Tracer project and record the output.

SAMPLE ADDRESSING PLAN:

Device	Interface / Role	IP address / pool
Wireless router	LAN / default gateway	192.168.10.1/24
DHCP server (optional)	Static LAN address	192.168.10.2/24
DHCP clients	Automatic configuration	192.168.10.100–192.168.10.149
DNS server	DHCP option	Use lab DNS / upstream DNS
WAN / Internet cloud	Upstream side	Depends on configured lab topology

OUTPUT:

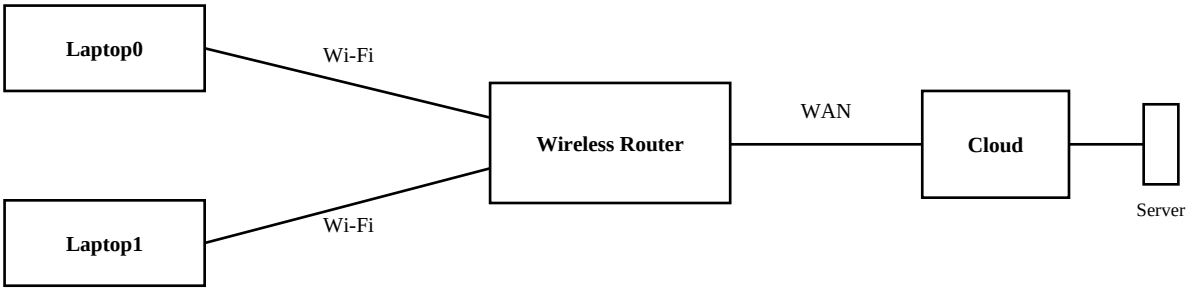

Fig. 1. Example internetwork topology with wireless clients, wireless router and Internet cloud.

CONFIGURATION AND SAMPLE OUTPUT

WIRELESS ROUTER CONFIGURATION:

In Packet Tracer, open the wireless router's GUI/Config tab. Set the LAN IP address, enable DHCP, configure the address pool, set the SSID to LAB-WIFI and choose WPA2-PSK security. The exact menu names vary by router model.

SAMPLE DHCP CLIENT OUTPUT:

```
PC> ipconfig

Wireless0 Connection:
   IPv4 Address........: 192.168.10.100
   Subnet Mask.........: 255.255.255.0
   Default Gateway.....: 192.168.10.1
   DNS Server..........: 192.168.10.2
```

SAMPLE CONNECTIVITY TEST:

```
PC> ping 192.168.10.1

Pinging 192.168.10.1 with 32 bytes of data:
Reply from 192.168.10.1: bytes=32 time<1ms TTL=64
Reply from 192.168.10.1: bytes=32 time<1ms TTL=64
Reply from 192.168.10.1: bytes=32 time<1ms TTL=64
Reply from 192.168.10.1: bytes=32 time<1ms TTL=64

Ping statistics:
   Sent = 4, Received = 4, Lost = 0 (0% loss)
```

SAMPLE DHCP SERVER POOL:

```
Pool Name       : WIFI_POOL
Default Gateway : 192.168.10.1
DNS Server      : 192.168.10.2
Start IP        : 192.168.10.100
Subnet Mask     : 255.255.255.0
Maximum Users   : 50
```

OBSERVATION:

1. Wireless clients associate with the configured SSID when the correct security key is entered.

2. DHCP automatically assigns a unique IP address and the relevant network parameters to each client.

3. A successful ping to 192.168.10.1 confirms connectivity between a wireless client and the router's LAN interface.

4. Internet/cloud connectivity depends on the upstream network, WAN addressing, routing and the cloud's configured connections.

5. In Simulation mode, DHCP Discover, Offer, Request and Acknowledge packets can be inspected when supported by the selected devices.

RESULT:

Thus, an internetwork using a wireless router, DHCP service and an Internet cloud was designed in Cisco Packet Tracer. Wireless clients were configured to obtain network settings automatically, and local connectivity was tested. External connectivity was considered according to the configured upstream network.

Note: The output shown is illustrative sample output, not a screenshot of a live Packet Tracer run. Replace it with screenshots from your own simulation if required by your lab instructor.